

Natural Language Processing

Lab 5: Topic modeling scikit-learn, pyLDAvis and BERTopic

2024/25

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1. For this week's lab session we will follow the example based on real world datasets "Topic extraction with Non-negative Matrix Factorization and Latent Dirichlet Allocation" from the scikit-learn documentation
 - (a) Follow the tutorial here: https://scikit-learn.org/stable/auto_examples/applications/plot_topics_extraction_with_nmf_lda.html
 - (b) The tutorial provides Jupyter notebook and source code files. Download whichever you prefer, and step through the code.
 - (c) First, install any additional dependencies that are needed and ensure that the script runs from start to finish. If it runs correctly, several bar charts showing word-topic distributions should appear.
 - (d) Experiment with different parameters for the vectorizer and LDA algorithms. How do they affect the topics produced?
 - (e) Make sure you understand what the vectorizer methods are doing. Read the documentation, try out extreme values for the parameters and inspect the results.
 - (f) Which vectorizer is used with which algorithm? Can you find out why?
 2.
 - (a) Read the documentation for pyLDAvis and import the model fitted by the script in question 1 for visualisation. <https://pypi.org/project/pyLDAvis/>
 3. In the lecture we also discussed the BERTopic package for topic modeling. Extensive documentation and example code is available on the homepage: <https://maartengr.github.io/BERTopic/>.
 - (a) We have used two text corpora so far in the lab sessions: the recipes corpus from week 2, and the inaugural speeches corpus from week 3. Read the documentation and guides provided, and train a topic model on both of these corpora.
 - (b) How does BERTopic choose the number of topics?
 - (c) Which techniques in the BERTopic pipeline gave you the best results on the recipes and speeches collections?